

Practice Quiz: Action

Part A: Multiple Choice

1. In Active Inference, each crochet stitch is best understood as: A) A random event with no predictable outcome B) A policy execution — a coordinated motor sequence that transforms the fabric C) A passive observation of the yarn D) An external state imposed on the agent
2. When a crocheter inserts the hook under the front loop only (instead of both loops), they are: A) Making an error that must be corrected B) Selecting a different motor sub-action that produces a different fabric texture C) Destroying the Markov blanket D) Ignoring their generative model
3. Frogging (ripping back stitches to correct errors) is best described as: A) A failure of the agent B) Active inference — corrective action that changes the world to reduce prediction error C) A random process D) Perceptual inference only
4. The range of stitches a crocheter knows (sc, dc, hdc, treble, etc.) constitutes their: A) Markov blanket B) External states C) Policy repertoire — the set of available actions for shaping the fabric D) Internal states
5. Working two stitches into the same space (an increase) is an example of: A) A shaping action that changes the geometry of the fabric to match the predicted form B) A perceptual error C) An external state change D) The generative model dissolving
6. When a crocheter decides whether to frog or continue despite an error, they are: A) Not performing any inference B) Evaluating competing policies based on expected free energy C) Ignoring the error completely D) Maximizing surprise
7. The pull-through force applied during each stitch affects fabric tightness. This force is: A) Irrelevant to the action B) A continuous motor variable the agent calibrates based on sensory feedback C) Always the same regardless of yarn or pattern D) An external state the crocheter cannot control

Part B: Short Answer

1. Describe the complete motor policy for a double crochet stitch, step by step. How does each sub-action transform the state of the system?
2. A crocheter discovers an error 5 rows back. They could frog all 5 rows, or they could continue and try to compensate. Describe the expected free energy trade-off of each option.
3. Explain how learning a new stitch (such as a post stitch or bobble) expands the crocheter's policy repertoire and enables new fabric outcomes that were previously impossible.